

AIM

To capture, filter and analyze network packets using the Wireshark packet sniffing tool.

REQUIREMENTS

- Wireshark
- Computer with network connection
- Ethernet/Wi-Fi network interface

THEORY

Wireshark is a network protocol analyzer used to capture and examine network packets in real time. It displays packet information such as source, destination, protocol, length and protocol fields.

A packet sniffer is a passive program that captures copies of packets transmitted or received by a computer. It does not generate packets itself.

Wireshark provides:

- Real-time packet capture
- Protocol decoding
- Display and capture filters
- Packet details and hexadecimal data
- Flow graph and statistics
- Network troubleshooting and analysis

PROCEDURE

1. Open **Wireshark**.
2. Select the active network interface such as **Ethernet** or **Wi-Fi**.
3. Start packet capture.
4. Allow packets to be captured.
5. Stop the capture after approximately **100 packets**.
6. Save the captured packets.
7. Apply suitable display filters to analyze different protocols.
8. Select individual packets to view their **Packet List, Packet Details and Packet Bytes**.

FILTERS USED

Experiment	Wireshark Display Filter
TCP	<code>tcp</code>
UDP	<code>udp</code>
ARP	<code>arp</code>
DNS	<code>dns</code>
HTTP	<code>http</code>
IP	<code>ip</code>
ICMP	<code>icmp</code>
DHCP	<code>dhcp</code>

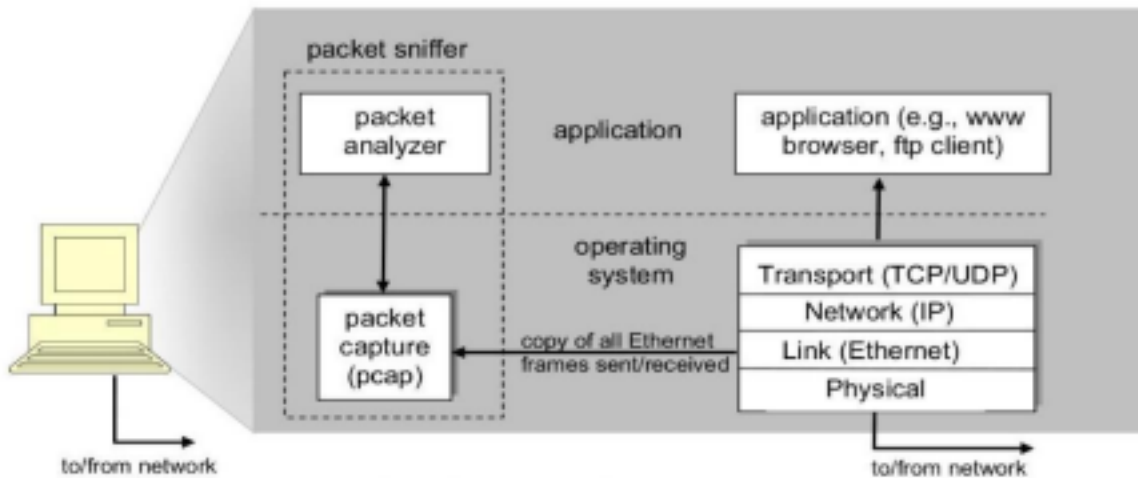
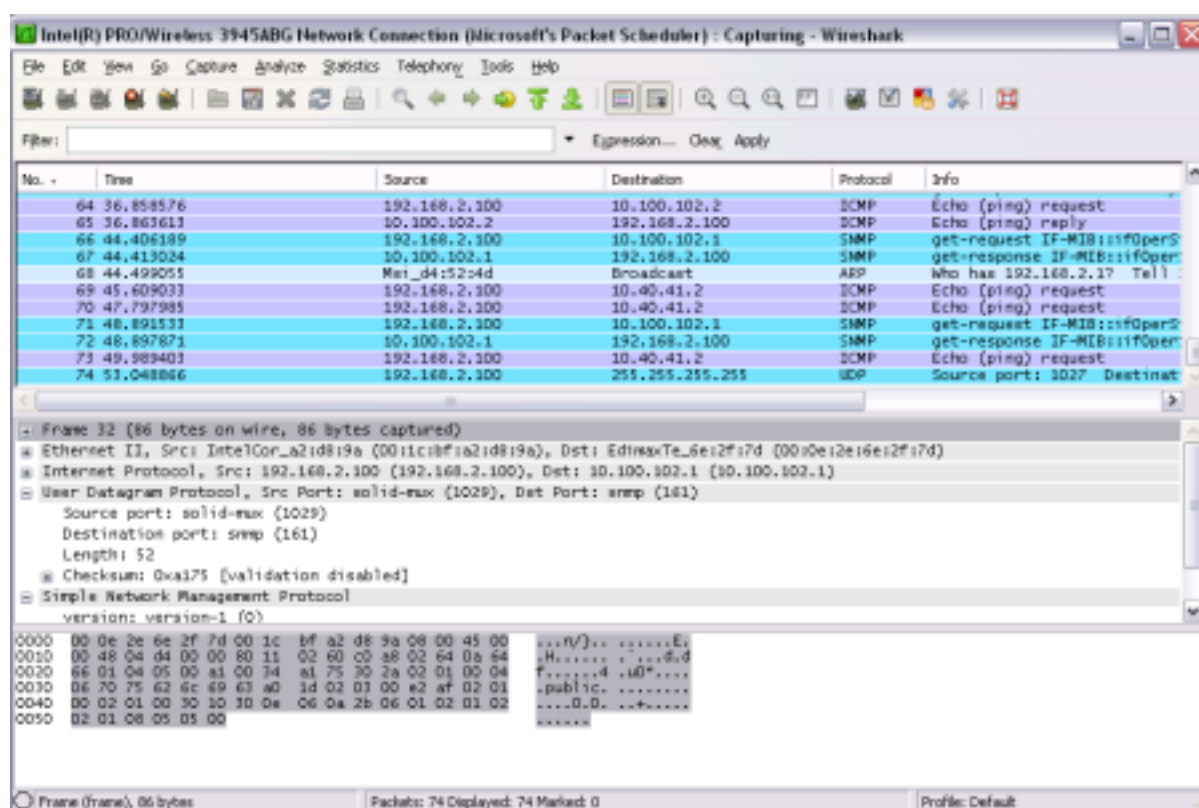
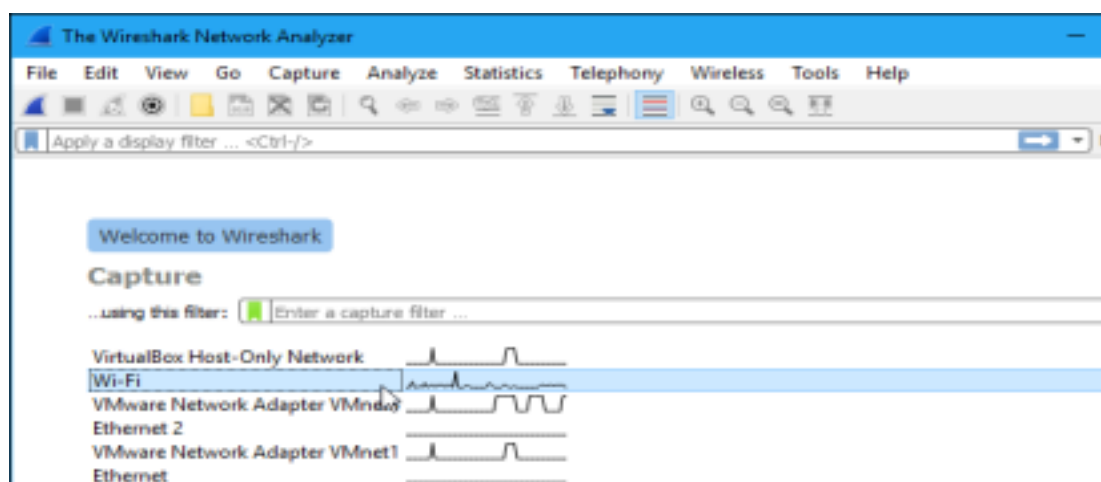
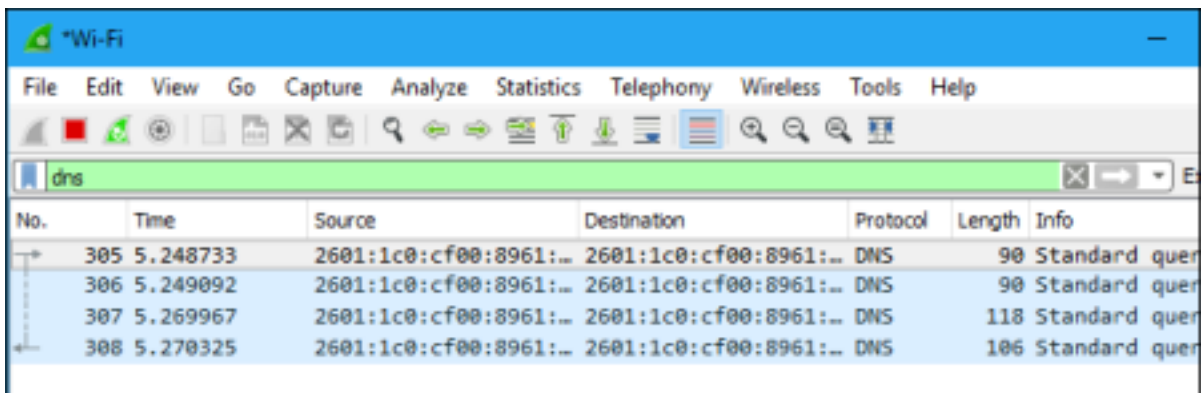
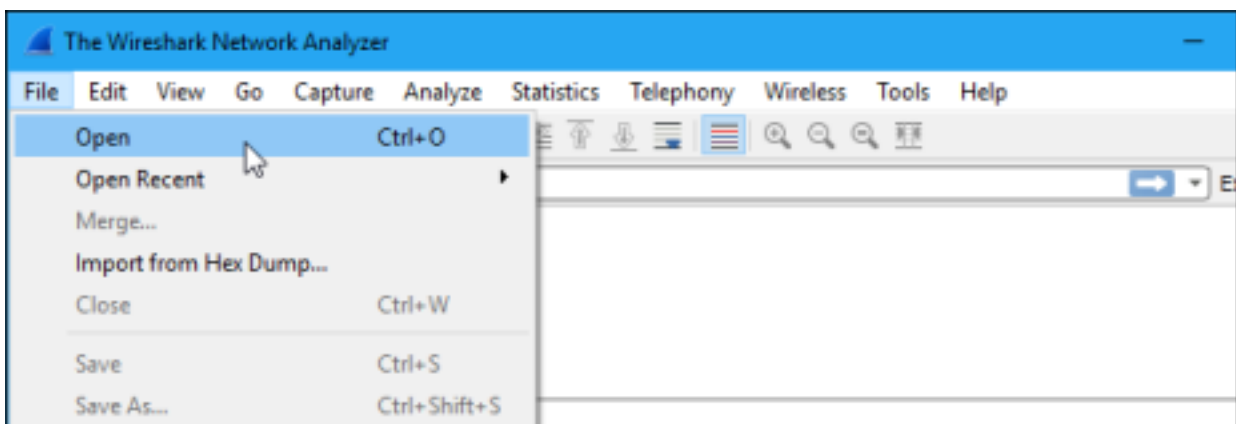
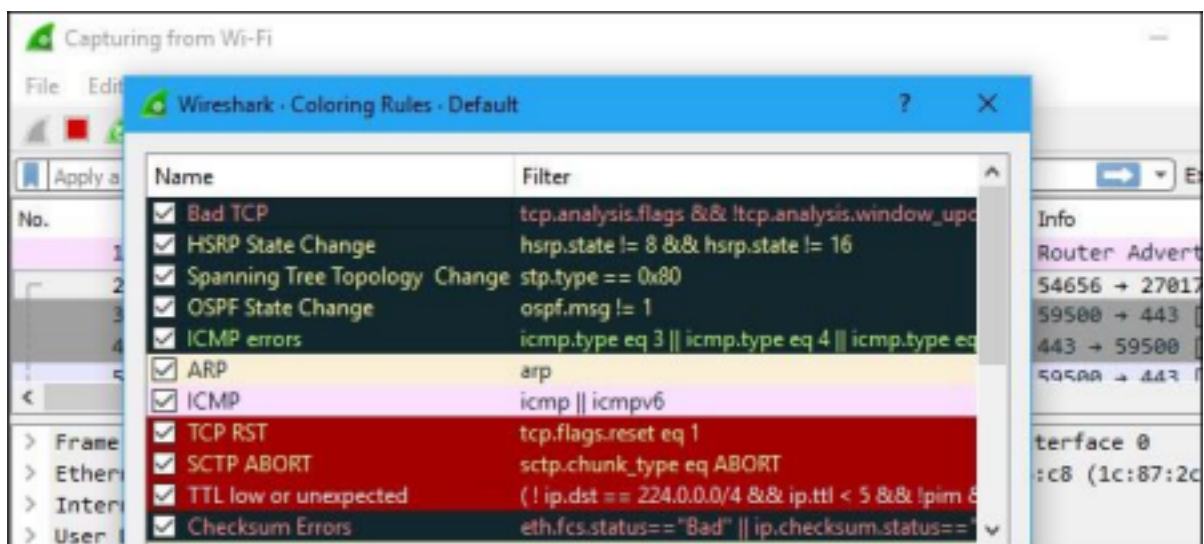
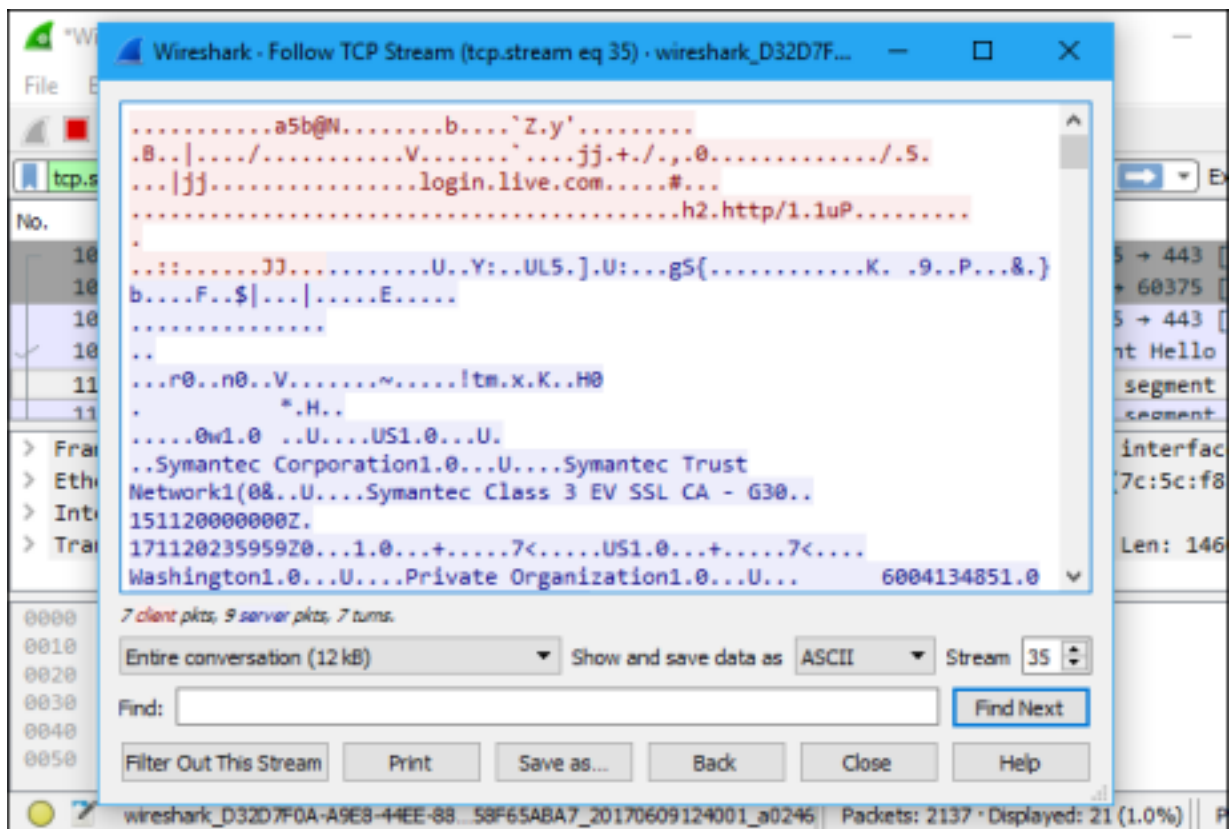
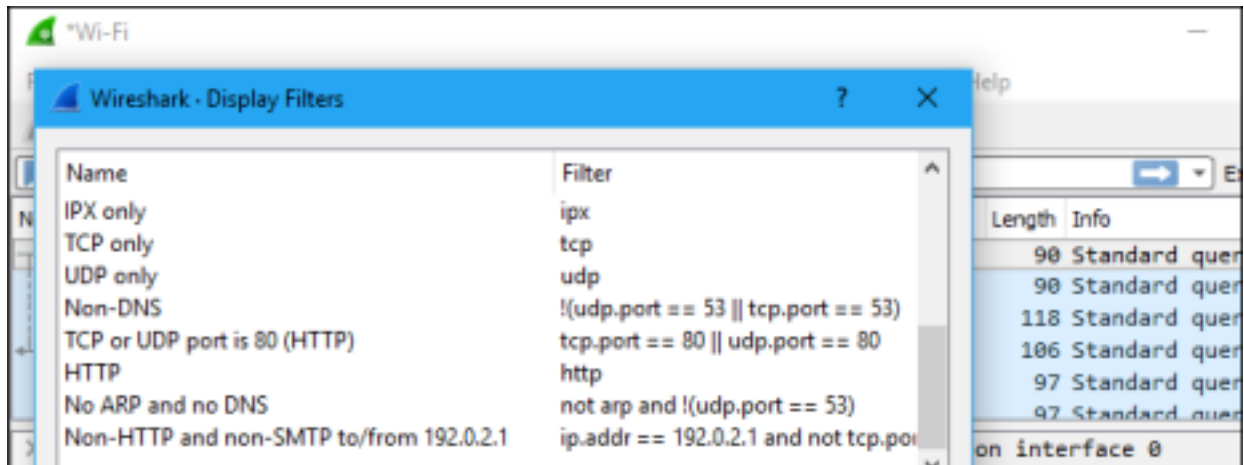
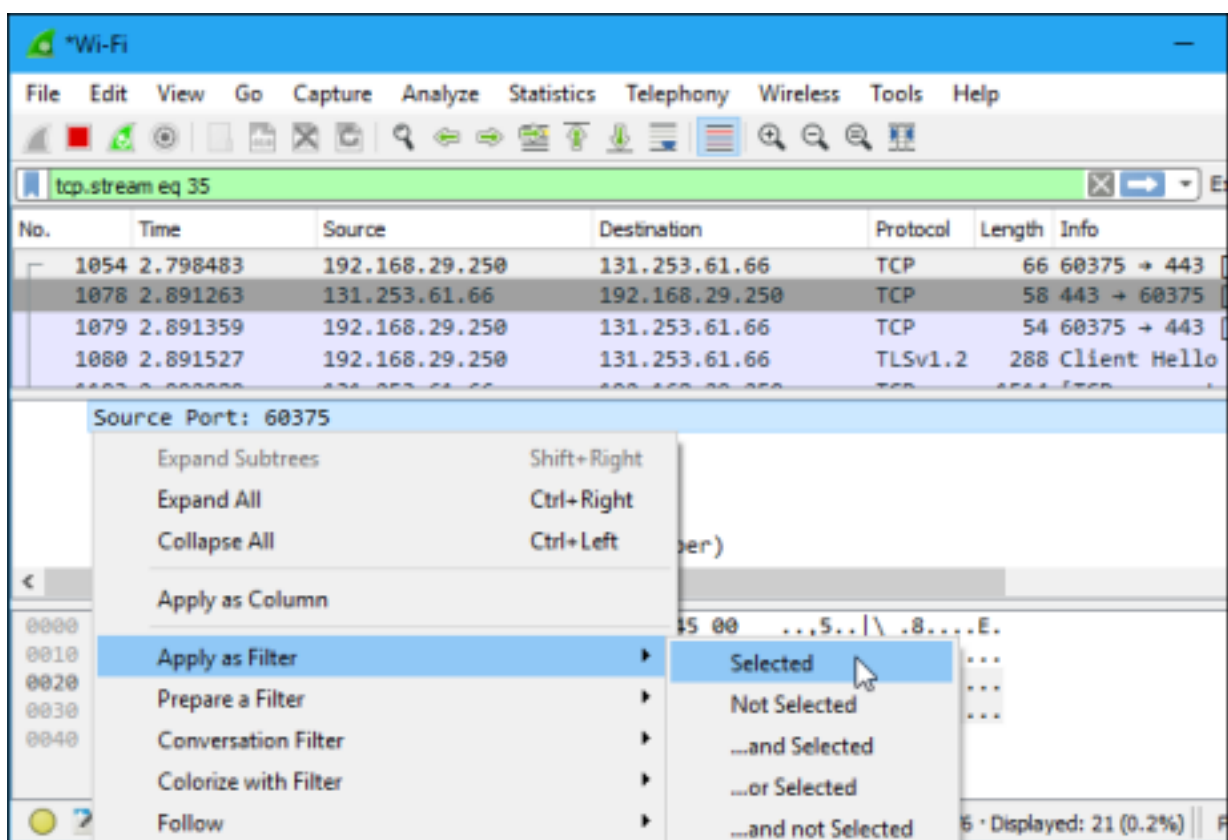
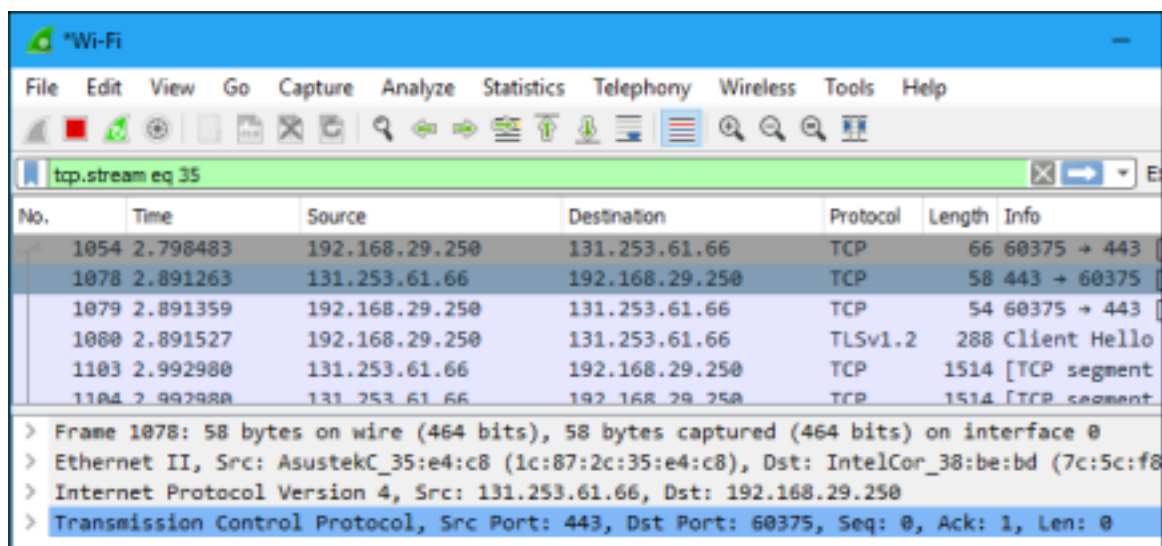


Figure 1: Packet sniffer structure









PACKET ANALYSIS

1. TCP/UDP Packets

Apply the filter `tcp` or `udp` in the display filter bar. Select a packet and inspect its source IP, destination IP, source port, destination port and protocol information.

The flow graph can be viewed using **Statistics** → **Flow Graph**.

2. ARP Packets

Apply the filter `arp`. ARP packets are used to map an IP address to a MAC address. The packet details show the sender and target IP/MAC addresses.

3. DNS Packets

Apply the filter `dns`. DNS packets contain information about domain-name queries and responses. The flow graph can be used to observe communication between the client and DNS server.

4. HTTP Packets

Apply the filter `http`. The captured packets can be inspected to identify HTTP requests, responses, source and destination addresses, and other HTTP fields.

5. IP/ICMP Packets

Apply `ip` or `icmp`. IP packets show source and destination IP addresses, while ICMP packets are commonly generated by tools such as `ping`.

6. DHCP Packets

Apply the filter `dhcp`. DHCP packets can be inspected to observe the process of obtaining network configuration information such as an IP address.

RESULT

Packets were successfully captured and analyzed using Wireshark. TCP/UDP, ARP, DNS, HTTP, IP/ICMP and DHCP packets were filtered and their protocol fields were inspected. Flow graphs were also generated for the required packets.

CONCLUSION

Thus, network packets were successfully captured, filtered and analyzed using the **Wireshark packet capture tool**.